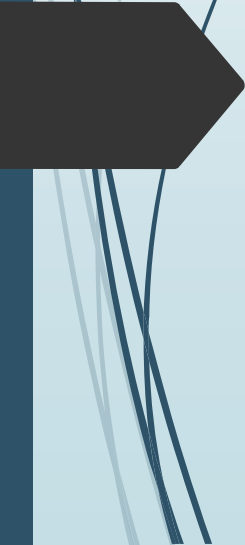


Introduction

Chapter 1.



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Review of Taylor Series



Section 1.2

youtube.com/watch?v=3d6DsjlBzJ4

Review of Taylor Series

- Most students will have encountered infinite series (particularly Taylor series) in their study of calculus without necessarily having acquired a good understanding of this topic.
- Once students are understanding of Taylor series, the Mean-Value Theorem, and alternating series (all topics in this section) as well as computer number representation (Section 2.2), they can proceed to study the fundamentals of numerical methods with better comprehension.

Taylor Series

- Familiar (and useful) examples of Taylor series are the following:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad (|x| < \infty) \quad (1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \quad (|x| < \infty) \quad (2)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} \quad (|x| < \infty) \quad (3)$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{k=0}^{\infty} x^k \quad (|x| < 1) \quad (4)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k} \quad (-1 < x \leq 1) \quad (5)$$

- Series (1)–(5) are Taylor series expanded about $c = 0$.

Taylor Series

- Taylor series expanded about $c = 1$:

$$\ln(x) = (x - 1) - \frac{(x - 1)^2}{2} + \frac{(x - 1)^3}{3} - \dots = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{(x - 1)^k}{k} \quad : 0 < x \leq 2.$$

- Series of this type are often used to compute good approximate values of complicated functions at specific points.
- **Example1** Use five terms in Series (5) to approximate $\ln(1.1)$, $S \approx 0.0953103333\dots$
- **Example2** Try to compute e^8 by using Series (1).

$$e^8 = 1 + 8 + \frac{64}{2} + \frac{512}{6} + \frac{4096}{24} + \frac{32768}{120} + \dots$$

- It is apparent that many terms will be needed to compute e^8 with reasonable precision. By repeated squaring, we find $e^2 = 7.389056$, $e^4 = 54.5981500$, and $e^8 = 2980.957987$. The first six terms given above yield 570.06666 5.

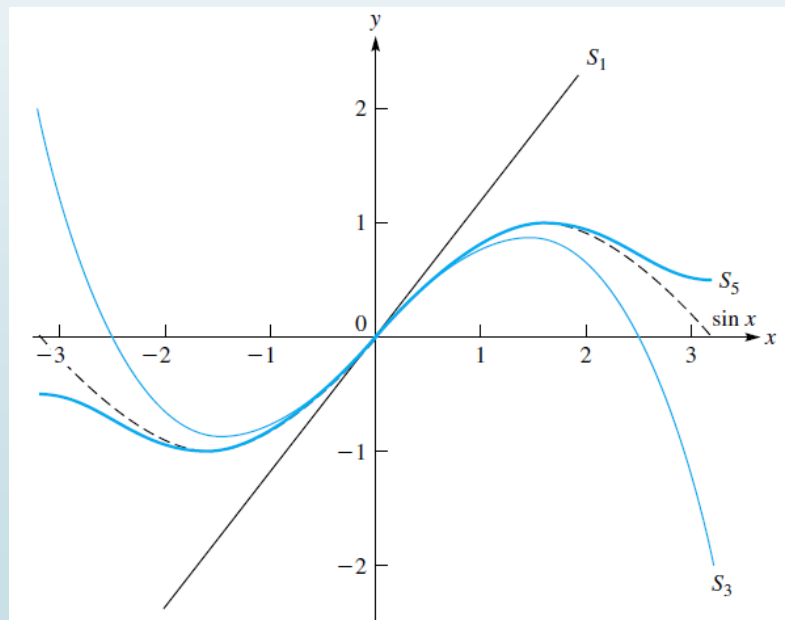
Taylor Series

- These examples illustrate a general rule:
A Taylor series converges rapidly near the point of expansion and slowly (or not at all) at more remote points.
- In the following Figure, we show the function $y = \sin x$ and the partial-sum functions S_1 , S_3 , S_5 which come from (Series (2)):

$$S_1 = x$$

$$S_3 = x - \frac{x^3}{6}$$

$$S_5 = x - \frac{x^3}{6} + \frac{x^5}{120}$$



Taylor Series

- All of the series illustrated above are examples of the following general series:

➤ Theorem 1 :

FORMAL TAYLOR SERIES FOR f ABOUT c

$$f(x) \sim f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \frac{f'''(c)}{3!}(x - c)^3 + \dots$$

$$f(x) \sim \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x - c)^k \quad (6)$$

- Series (6) is called the “**Taylor series of f at the point c .**”
- In the special case $c = 0$, Series (6) is also called a **Maclaurin series**:

$$f(x) \sim f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

$$f(x) \sim \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k \quad (7)$$

Taylor Series

- **Example3** What is the Taylor series of the function: $f(x) = 3x^5 - 2x^4 + 15x^3 + 13x^2 - 12x - 5$ at the point $c = 2$?
- **Solution:** To compute the coefficients in the series, we need the numerical values of $f^{(k)}(2)$ for $k \geq 0$.
- For $k \geq 6$. Therefore, we have :

$$f(x) \sim 207 + 396(x - 2) + 295(x - 2)^2 + 119(x - 2)^3 + 28(x - 2)^4 + 3(x - 2)^5$$

- In this example, it is not difficult to see that $\sim$ may be replaced by $=$.?

Complete Horner's Algorithm:

- An application of Horner's algorithm is that of finding the Taylor expansion of a polynomial about any point.
- Let $p(x)$ be a given polynomial of degree n with coefficients a_k

$$\begin{aligned} p(x) &= a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 \\ &= c_n (x - r)^n + c_{n-1} (x - r)^{n-1} + \dots + c_1 (x - r) + c_0 \end{aligned}$$

- Of course, Taylor's Theorem asserts that $c_k = p^{(k)}(r)/k!$, notice that $p(r) = c_0$, so this coefficient is obtained by applying Horner's algorithm to the polynomial p with the point r .
- The algorithm also yields the polynomial:

$$q(x) = \frac{p(x) - p(r)}{x - r} = c_n (x - r)^{n-1} + c_{n-1} (x - r)^{n-2} + \dots + c_1$$

Complete Horner's Algorithm:

- This shows that the second coefficient, c_1 , can be obtained by applying Horner's algorithm to the polynomial q with point r , because $c_1 = q(r)$.
- This process is repeated until all coefficients c_k are found. We call the algorithm just described the **complete Horner's algorithm**.
- The pseudocode for executing it is arranged so that the coefficients c_k overwrite the input coefficients a_k .

```
integer  $n, k, j$ ;  real  $r$ ;  real array  $(a_i)_{0:n}$   
for  $k = 0$  to  $n - 1$  do  
    for  $j = n - 1$  to  $k$  do  
         $a_j \leftarrow a_j + r a_{j+1}$   
    end for  
end for
```

Continue →

Complete Horner's Algorithm:

- This procedure can be used in carrying out **Newton's method** for finding roots of a polynomial, which we discuss in Chapter 3.
- **Example 4** Using the complete Horner's algorithm, find the Taylor expansion of the polynomial:
 $p(x) = x^4 - 4x^3 + 7x^2 - 5x + 2$ about the point $r = 3$.
- **Solution:** The work can be arranged as follows:

$$\begin{array}{r}
 \begin{array}{rrrrr}
 1 & -4 & 7 & -5 & 2 \\
 3) & & 3 & -3 & 12 & 21 \\
 \hline
 1 & -1 & 4 & 7 & 23 \\
 & 3 & 6 & 30 & \\
 \hline
 1 & 2 & 10 & 37 & \\
 & 3 & 15 & & \\
 \hline
 1 & 5 & 25 & & \\
 & 3 & & & \\
 \hline
 1 & 8 & & &
 \end{array}
 \end{array}$$

The calculation shows that

$$p(x) = (x - 3)^4 + 8(x - 3)^3 + 25(x - 3)^2 + 37(x - 3) + 23$$

Taylor Theorem in Terms of $(x-c)$?

► THEOREM 2

TAYLOR'S THEOREM FOR $f(x)$

If the function f possesses continuous derivatives of orders $0, 1, 2, \dots, (n + 1)$ in a closed interval $I = [a, b]$, then for any c and x in I ,

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x - c)^k + E_{n+1} \quad (8)$$

where the error term E_{n+1} can be given in the form

$$E_{n+1} = \frac{f^{(n+1)}(\xi)}{(n + 1)!} (x - c)^{n+1}$$

Here ξ is a point that lies between c and x and depends on both.

- In practical computations with Taylor series, it is usually necessary to *truncate* the series. A series is said to be **truncated** if we ignore all terms after a certain point. Thus, if we truncate the exponential Series (1) after seven terms, the result is:

Taylor Theorem in Terms of $(x-c)$?

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$$

- This no longer represents e^x except when $x = 0$. But the truncated series should *approximate* e^x .
- Here is where we need Taylor's Theorem. With its help, we can assess the difference between a function f and its truncated Taylor series.

Taylor Theorem in Terms of (x-c)?

- **Example 5** Derive the Taylor series for e^x at $c = 0$, and prove that it converges to e^x by using Taylor's Theorem.
- **Solution:** If $f(x) = e^x$, then $f^{(k)}(x) = e^x$ for $k \geq 0$. Therefore, $f^{(k)}(c) = f^{(k)}(0) = e^0 = 1$ for all k .
- From Equation (8), we have :

$$e^x = \sum_{k=0}^n \frac{x^k}{k!} + \frac{e^{\xi}}{(n+1)!} x^{n+1} \quad (9)$$

- Now let us consider all the values of x in some symmetric interval around the origin, for example, $-s \leq x \leq s$. Then $|x| \leq s$, $|\xi| \leq s$, and $e^{\xi} \leq e^s$. Hence, the remainder term satisfies this inequality:

$$\lim_{n \rightarrow \infty} \left| \frac{e^{\xi}}{(n+1)!} x^{n+1} \right| \leq \lim_{n \rightarrow \infty} \frac{e^s}{(n+1)!} s^{n+1} = 0$$

Taylor Theorem in Terms of (x-c)?

- Thus, if we take the limit as $n \rightarrow \infty$ on both sides of Equation (9), we obtain :

$$e^x = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{x^k}{k!} = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

- This example illustrates how we can establish, in specific cases, that a formal Taylor Series (9) actually represents the function.
- Let's examine another example to see how the formal series can **fail** to represent the function.

Taylor Theorem in Terms of $(x-c)$?

► **Example 6** Derive the formal Taylor series for $f(x) = \ln(1+x)$ at $c = 0$, and determine the range of positive x for which the series represents the function.

► **Solution:** We need $f^{(k)}(x)$ and $f^{(k)}(0)$ for $k \geq 1$.

$$f(x) = \ln(1+x)$$

$$f(0) = 0$$

$$f'(x) = (1+x)^{-1}$$

$$f'(0) = 1$$

$$f''(x) = -(1+x)^{-2}$$

$$f''(0) = -1$$

$$f'''(x) = 2(1+x)^{-3}$$

$$f'''(0) = 2$$

$$f^{(4)}(x) = -6(1+x)^{-4}$$

$$f^{(4)}(0) = -6$$

$$\vdots$$

$$\vdots$$

$$f^{(k)}(x) = (-1)^{k-1}(k-1)!(1+x)^{-k}$$

$$f^{(k)}(0) = (-1)^{k-1}(k-1)!$$

Taylor Theorem in Terms of (x-c)?

- Hence, by Taylor Theorem we obtain:

$$\begin{aligned}\ln(1+x) &= \sum_{k=1}^n (-1)^{k-1} \frac{(k-1)!}{k!} x^k + \frac{(-1)^n n! (1+\xi)^{-n-1}}{(n+1)!} x^{n+1} \\ &= \sum_{k=1}^n (-1)^{k-1} \frac{x^k}{k} + \frac{(-1)^n}{n+1} (1+\xi)^{-n-1} x^{n+1}\end{aligned}\tag{10}$$

- For the **infinite** series to represent $\ln(1+x)$, it is necessary and sufficient that the error term converge to zero as $n \rightarrow \infty$.
- Assume that $0 \leq x \leq 1$. Then $0 \leq \xi \leq x$ (because zero is the point of expansion); thus, $0 \leq x/(1+\xi) \leq 1$. Hence, the error term converges to zero in this case. If $x > 1$, the terms in the series do not approach zero, and the series does not converge.

Mean Value Theorem?

- The special case $n = 0$ in Taylor's Theorem is known as the **Mean-Value Theorem**.

- **Theorem3.**

MEAN-VALUE THEOREM

If f is a continuous function on the closed interval $[a, b]$ and possesses a derivative at each point of the open interval (a, b) , then

$$f(b) = f(a) + (b - a)f'(\xi)$$

for some ξ in (a, b) .

- That is, for some $\xi \in (a, b)$,

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

- The right-hand side could be used as an *approximation* for $f'(x)$ at any x within the interval (a, b) . The approximation of derivatives is discussed more fully in Section 4.3.

Taylor's Theorem in Terms of h

- Other forms of Taylor's Theorem are often useful. These can be obtained from the basic formula (8) by changing the variables.

COROLLARY 1

TAYLOR'S THEOREM FOR $f(x + h)$

If the function f possesses continuous derivatives of order $0, 1, 2, \dots, (n + 1)$ in a closed interval $I = [a, b]$, then for any x in I ,

$$f(x + h) = \sum_{k=0}^n \frac{f^{(k)}(x)}{k!} h^k + E_{n+1} \quad (11)$$

where h is any value such that $x + h$ is in I and where

$$E_{n+1} = \frac{f^{(n+1)}(\xi)}{(n + 1)!} h^{n+1}$$

for some ξ between x and $x + h$.

- The form (11) is obtained from Equation (8) by replacing x by $x + h$ and replacing c by x .

Taylor's Theorem in Terms of h

- The **error term** E_{n+1} depends on h in two ways: First, h^{n+1} is explicitly present; second, the point ξ generally depends on h .
- To express this qualitative fact, we write:

$$E_{n+1} = \mathcal{O}(h^{n+1})$$

as $h \rightarrow 0$. This is called **big O notation**, and it is shorthand for the inequality :

$$|E_{n+1}| \leq C|h|^{n+1}$$

where C is a constant.

Taylor Theorem in Terms of h ?

- **Example 7** Expand $\sqrt{1+h}$ in powers of h . Then compute $\sqrt{1.00001}$ and $\sqrt{0.99999}$.
- **Solution:** Let $f(x) = x^{1/2}$. Then $f'(x) = \frac{1}{2}x^{-1/2}$, $f''(x) = -\frac{1}{4}x^{-3/2}$, $f'''(x) = \frac{3}{8}x^{-5/2}$, and so on. Now use Equation (11) with $x = 1$. Taking $n = 2$ for illustration, we have:

$$\sqrt{1+h} = 1 + \frac{1}{2}h - \frac{1}{8}h^2 + \frac{1}{16}h^3\xi^{-5/2} \quad (12)$$

- where ξ is an unknown number that satisfies $1 < \xi < 1+h$, if $h > 0$. It is important to notice that the function $f(x) = \sqrt{x}$ possesses derivatives of all orders at any point $x > 0$.
- In Equation (12), let $h = 10^{-5}$. Then

$$\sqrt{1.00001} \approx 1 + 0.5 \times 10^{-5} - 0.125 \times 10^{-10} = 1.000004999987500$$

- By substituting $-h$ for h in the series (12), we obtain:

$$\sqrt{1-h} = 1 - \frac{1}{2}h - \frac{1}{8}h^2 - \frac{1}{16}h^3\xi^{-5/2}$$

- Hence, we have:

$$\sqrt{0.99999} \approx 0.999994999987500$$

- Since $1 < \xi < 1+h$, the absolute error does not exceed

$$\frac{1}{16}h^3\xi^{-5/2} < \frac{1}{16}10^{-15} = 0.000000000000000625$$

- and both numerical values are correct to all 15 decimal places shown.

Alternating Series

- This theorem applies only to **alternating series** - that is, series in which the successive terms are alternately positive and negative.

► Theorem 4

ALTERNATING SERIES THEOREM

If $a_1 \geq a_2 \geq \cdots \geq a_n \geq \cdots 0$ for all n and $\lim_{n \rightarrow \infty} a_n = 0$, then the alternating series

$$a_1 - a_2 + a_3 - a_4 + \cdots$$

converges; that is,

$$\sum_{k=1}^{\infty} (-1)^{k-1} a_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n (-1)^{k-1} a_k = \lim_{n \rightarrow \infty} S_n = S$$

where S is its sum and S_n is the n th partial sum. Moreover, for all n ,

$$|S - S_n| \leq a_{n+1}$$

- **Example 8.** If the sin series is to be used in computing $\sin 1$ with an error less than $\frac{1}{2} \times 10^{-6}$, how many terms are needed?

Alternating Series

- **Solution:** From Series (2), we have

$$\sin 1 = 1 - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} + \dots$$

- If we stop at $1/(2n-1)!$, the error does not exceed the first neglected term, which is $1/(2n+1)!$. Thus, we should select n so that

$$\frac{1}{(2n+1)!} < \frac{1}{2} \times 10^{-6}$$

- Using logarithms to base 10, we obtain

$$\log(2n+1)! > \log 2 + 6 = 6.3.$$

With a calculator, we compute a table of values for $\log n!$ and find that $\log 10! \approx 6.6$. Hence, if $n = 5$, the error will be acceptable.

- (1) The **Taylor series expansion** about c for $f(x)$ is

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x - c)^k + E_{n+1}$$

- with error term

$$E_{n+1} = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - c)^{n+1}$$

- A more useful form for us is the **Taylor series expansion** for $f(x + h)$, which is

$$f(x + h) = \sum_{k=0}^n \frac{f^{(k)}(x)}{k!} h^k + E_{n+1}$$

- with error term

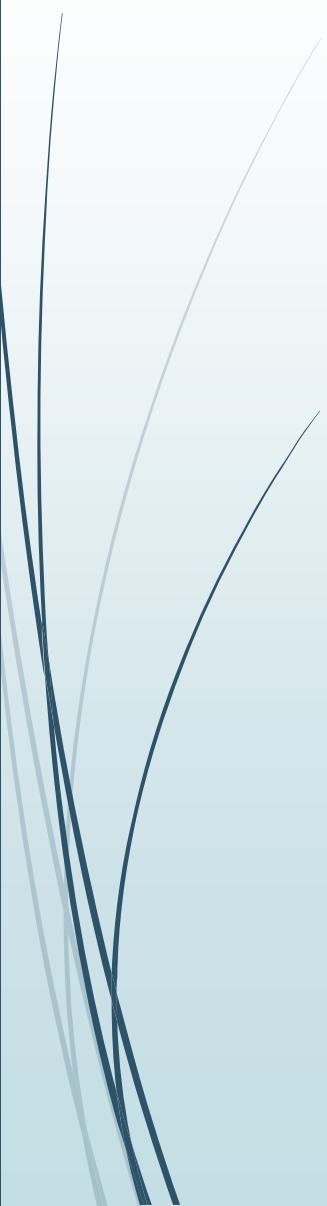
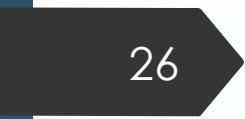
$$E_{n+1} = \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1} = \mathcal{O}(h^{n+1})$$

► (2) An **alternating series**

$$S = \sum_{k=1}^{\infty} (-1)^{k-1} a_k$$

- converges when the terms a_k converge downward to zero. Furthermore, the partial sums S_n differ from S by an amount that is bounded by

$$|S - S_n| \leq a_{n+1}$$



END